Paper OOL-6: Autocatalytic Closure and Chemical Memory: When Reaction Networks Begin to Remember

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper isolates autocatalytic closure from later claims about alphabets, homochirality, or protocells. A reaction network begins to remember only when its products help regenerate the conditions that make them. In CDFD terms, this is the first serious biological use of M_s: retained organization through feedback. The companion script performs a conservative closure scan and reports sub-ignition persistence enhancement rather than completed metabolic ignition.

Symbol Conventions

CDFD origins-of-life symbol convention.

Symbol	Meaning in Part II	Origins-of-life reading
Φ	Available driving flux	Redox, thermal, photochemical, proton/ion, or chemical potential drive supplied by the environment
C	Effective constraint or resistance	Mineral geometry, pore confinement, diffusion barrier, permeability, activation barrier, or maintenance burden
S	Responsive organization	Surface, boundary, catalytic network, coacervate, or protocellular structure that routes flux into retained function
M_s	Retained structural memory	Composition, sequence, topology, compartment state, mineral imprint, or boundary history that persists across renewal cycles
Ψ_s	Local adaptive ratio $(\Phi/C) \cdot S \cdot M_s$	Bookkeeping gauge for whether drive, constraint, responsiveness, and retention are co-present; not a universal constant
Λ_{life}	Integrated Life Number where used	Capstone surplus ratio for decay, near-critical proto-biology, and sustained self-maintenance

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Defines how fast a constraint, pathway, or retained structure grows under load
β	Relaxation, decay, or washout coefficient	Defines recovery, loss, clearance, hydrolysis, or return toward baseline
γ	Coupling, diffusion, or redistribution coefficient	Defines spread across pores, surfaces, compartments, networks, or chemical phases
δ, ϵ	Perturbation, tolerance, or small offset scales	Used only as local thresholds, numerical guards, or experimental tolerances
η, λ, μ, ν	Efficiency, rate, baseline, or frequency terms	Meaning is set by the equation and subscript where they occur
ρ, σ, τ, ϕ	Density/correlation, conductance/noise, time-scale, or phase/field terms	Never universal constants unless a paper explicitly defines them that way
Ω, Λ	Domain/state space; Life Number where used	Ω names a modeled state space; Λ is reserved for life-number notation only where defined

1 Introduction

Papers 1–5 supplied energy, transport, compartments, and selective interfaces. None of these alone guarantees *self-reinforcing* chemistry. Autocatalysis and cross-catalysis supply positive feedback: products accelerate their own formation [1, 2]. In open systems, however, diffusion and dilution supply negative feedback. Closure is the balance point.

Gap from Paper 5: catalytic selection must escalate into network persistence.

Status: Inferred

Capstone channel mapping. Metabolic closure is not single-channel: losing input Φ, electron throughput σ_e, proton coupling σ_p, or stabilization capacity against S collapses the bundled J in the weakest-link sense even when Γ inequalities locally favor amplification.

2 Model and Assumptions

2.1 Network mass balances

For species X_i with concentrations c_i,

$$\frac{\partial c_i}{\partial t} = \sum_j R_{ij}(\boldsymbol{c}) - \nabla \cdot \boldsymbol{J}_i + S_i, \qquad \boldsymbol{J}_i = -\frac{1}{C_i}\nabla c_i, \tag{1}$$

with sources S_i. Here C_i is a *transport resistance* for species i (not to be confused with product concentration symbols in legacy drafts).

Status: Derived

2.2 Minimal autocatalytic core

$$X + A \to 2X, \qquad R = kc_X c_A. \tag{2}$$

Well-mixed loss to a compartment of scale L scales as $D_{\text{eff}} c_X / L^2$ with $D_{\text{eff}} \sim 1/C$ in the resistance picture. Then

$$\frac{dc_X}{dt} = kc_X c_A - \frac{1}{CL^2} c_X = c_X \left(kc_A - \frac{1}{CL^2} \right). \tag{3}$$

Status: Derived

2.3 Ignition parameter

Define

$$\Gamma \ = \ kc_A CL^2. \tag{4}$$

If $\Gamma > 1$, exponential growth of c_X results in this linearized well-mixed limit; if $\Gamma < 1$, trajectories decay toward zero without other sources. The boundary $\Gamma = 1$ is a mean-field caricature; spatial patterning and nonlinear saturation modify bifurcations.

Status: Derived

2.4 Vector closure

A network is *closed* if every required intermediate is produced internally faster than it leaks, $\forall i,\ \sum_j k_{ij} c_j > (C_i L_i^2)^{-1}$ in the same schematic sense.

Status: Hypothesis

2.5 Energy coupling

Rates may scale with coupled fluxes $R_i \propto |\boldsymbol{J}_e \cdot \boldsymbol{J}_p|$ when redox drive gates chemistry [3]. We do not specify constants.

Status: Hypothesis

2.6 Linear stability (Jacobian)

Let J_{jac} be the Jacobian of the reaction vector field. Local stability of a steady state requires $\operatorname{Re}\lambda < 0$ for all eigenvalues λ of J_{jac}. Autocatalysis can instead produce unstable modes, limit cycles, or blow-up depending on constraints and saturation.

Status: Derived

3 Main Results

3.1 Conceptual

Metabolic organization begins when positive feedback loops beat transport losses [4]. Compartments increase C (resistance to escape), improving Γ for fixed k, c_A, L.

Status: Inferred

4 Numerical Verification

`supplementary_ool_06.py` integrates a reduced two-parameter caricature. Reported final Φ values:

- $(\Phi_0 = 0.1, C = 1.0) \to 0.073$
- $(\Phi_0 = 0.5, C = 1.0) \to 0.554$
- $(\Phi_0 = 0.1, C = 5.0) \to 0.102$
- $(\Phi_0 = 0.5, C = 5.0) \to 0.849$

Higher constraint C raises the final state relative to comparable baselines, but **no run exhibits runaway ignition** in the companion script. Narratives about "critical metabolism" must therefore remain Hypothesis/Open until bifurcation scans identify supercritical parameters.

Status: Derived

5 Interpretation

The mathematics clarifies what must be proven: closure is a competition between amplification and loss. The numerics currently demonstrate *sub-ignition* persistence enhancement, not autonomous self-acceleration to a living steady state.

Status: Inferred

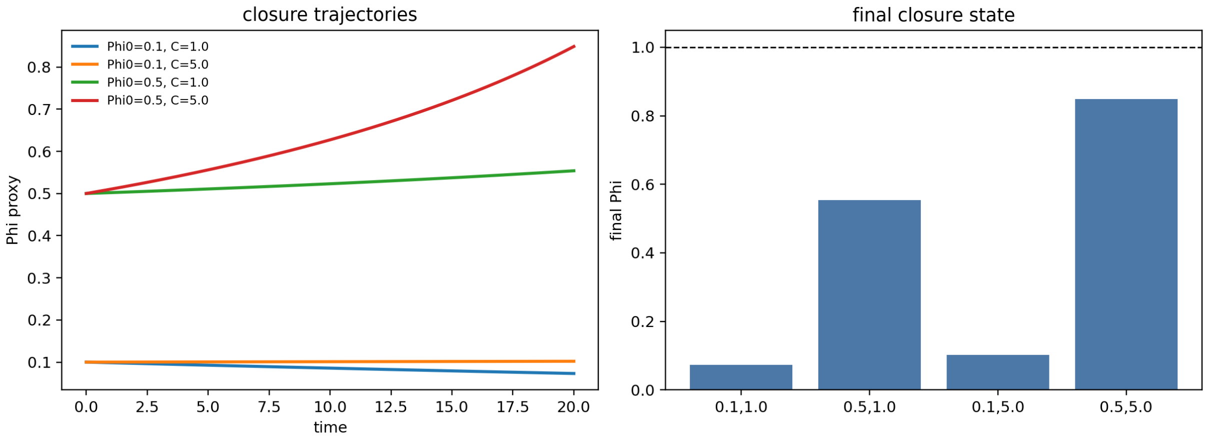

Figure 1: Autocatalytic-closure diagnostic. The trajectories show persistence enhancement under stronger constraint but do not prove completed metabolic ignition. The figure is a disciplined threshold check, not a claim of living metabolism.

6 Limitations and Open Problems

- Mismatch between rhetorical "ignition" and outputs is explicitly documented.
- Need bifurcation analysis in (k, c_A, C, L) space with spatial resolution.
- Saturation chemistry and side reactions omitted.

Status: Open

7 Conclusion

This paper states the closure criterion clearly and shows that the companion script does not yet realize full ignition. Paper 7 moves to aromatic carriers where pattern persistence becomes an information-relevant quantity.

Status: Inferred

8 Reproducibility Note

```
python supplementary_ool_06.py
```

9 Modern Network Boundary

Autocatalysis is not enough by itself. A closure claim must distinguish three layers: self-amplification of a species, mutual regeneration of a reaction set, and persistence of an organized network through environmental cycling. The modern CDFD reading is therefore strict: closure becomes biological only when it couples to chemical memory, boundary retention, and later copying. This is why Paper 6 ends before homochirality or genetic inheritance are claimed.

Status: Derived guardrail

References

[1] Manfred Eigen. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58:465–523, 1971.

[2] Stuart A. Kauffman. Autocatalytic sets of proteins. *Journal of Theoretical Biology*, 119:1–24, 1986.

[3] Nick Lane, John F. Allen, and William Martin. How did LUCA make a living? chemiosmosis in the origin of life. *BioEssays*, 32:271–280, 2010.

[4] John Maynard Smith and Eörs Szathmáry. *The Major Transitions in Evolution.* Oxford University Press, 1997.